

The following examples illustrate this.

Example 6.2.10

Show that $\mathbf{P}_3 = \text{span}\{x^2 + x^3, x, 2x^2 + 1, 3\}$.

Solution. Write $U = \text{span}\{x^2 + x^3, x, 2x^2 + 1, 3\}$. Then $U \subseteq \mathbf{P}_3$, and we use the fact that $\mathbf{P}_3 = \text{span}\{1, x, x^2, x^3\}$ to show that $\mathbf{P}_3 \subseteq U$. In fact, x and $1 = \frac{1}{3} \cdot 3$ clearly lie in U . But then successively,

$$x^2 = \frac{1}{2}[(2x^2 + 1) - 1] \quad \text{and} \quad x^3 = (x^2 + x^3) - x^2$$

also lie in U . Hence $\mathbf{P}_3 \subseteq U$ by Theorem 6.2.2.

Example 6.2.11

Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in a vector space V . Show that

$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \text{span}\{\mathbf{u} + 2\mathbf{v}, \mathbf{u} - \mathbf{v}\}$$

Solution. We have $\text{span}\{\mathbf{u} + 2\mathbf{v}, \mathbf{u} - \mathbf{v}\} \subseteq \text{span}\{\mathbf{u}, \mathbf{v}\}$ by Theorem 6.2.2 because both $\mathbf{u} + 2\mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ lie in $\text{span}\{\mathbf{u}, \mathbf{v}\}$. On the other hand,

$$\mathbf{u} = \frac{1}{3}(\mathbf{u} + 2\mathbf{v}) + \frac{2}{3}(\mathbf{u} - \mathbf{v}) \quad \text{and} \quad \mathbf{v} = \frac{1}{3}(\mathbf{u} + 2\mathbf{v}) - \frac{1}{3}(\mathbf{u} - \mathbf{v})$$

so $\text{span}\{\mathbf{u}, \mathbf{v}\} \subseteq \text{span}\{\mathbf{u} + 2\mathbf{v}, \mathbf{u} - \mathbf{v}\}$, again by Theorem 6.2.2.

Exercises for 6.2

Exercise 6.2.1 Which of the following are subspaces of $\mathbf{P}_3$? Support your answer.

- $U = \{f(x) \mid f(x) \in \mathbf{P}_3, f(2) = 1\}$
- $U = \{xg(x) \mid g(x) \in \mathbf{P}_2\}$
- $U = \{xg(x) \mid g(x) \in \mathbf{P}_3\}$
- $U = \{xg(x) + (1-x)h(x) \mid g(x) \text{ and } h(x) \in \mathbf{P}_2\}$
- $U =$ The set of all polynomials in $\mathbf{P}_3$ with constant term 0
- $U = \{f(x) \mid f(x) \in \mathbf{P}_3, \deg f(x) = 3\}$

Exercise 6.2.2 Which of the following are subspaces of $\mathbf{M}_{22}$? Support your answer.

- $U = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, \text{ and } c \text{ in } \mathbb{R} \right\}$
- $U = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a + b = c + d; a, b, c, d \text{ in } \mathbb{R} \right\}$
- $U = \{A \mid A \in \mathbf{M}_{22}, A = A^T\}$
- $U = \{A \mid A \in \mathbf{M}_{22}, AB = 0\}, B \text{ a fixed } 2 \times 2 \text{ matrix}$
- $U = \{A \mid A \in \mathbf{M}_{22}, A^2 = A\}$
- $U = \{A \mid A \in \mathbf{M}_{22}, A \text{ is not invertible}\}$

- g. $U = \{A \mid A \in \mathbf{M}_{22}, BAC = CAB\}$, B and C fixed 2×2 matrices

Exercise 6.2.3 Which of the following are subspaces of $\mathbf{F}[0, 1]$? Support your answer.

- $U = \{f \mid f(0) = 0\}$
- $U = \{f \mid f(0) = 1\}$
- $U = \{f \mid f(0) = f(1)\}$
- $U = \{f \mid f(x) \geq 0 \text{ for all } x \text{ in } [0, 1]\}$
- $U = \{f \mid f(x) = f(y) \text{ for all } x \text{ and } y \text{ in } [0, 1]\}$
- $U = \{f \mid f(x+y) = f(x) + f(y) \text{ for all } x \text{ and } y \text{ in } [0, 1]\}$
- $U = \{f \mid f \text{ is integrable and } \int_0^1 f(x)dx = 0\}$

Exercise 6.2.4 Let A be an $m \times n$ matrix. For which columns $\mathbf{b}$ in $\mathbb{R}^m$ is $U = \{\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n, A\mathbf{x} = \mathbf{b}\}$ a subspace of $\mathbb{R}^n$? Support your answer.

Exercise 6.2.5 Let $\mathbf{x}$ be a vector in $\mathbb{R}^n$ (written as a column), and define $U = \{A\mathbf{x} \mid A \in \mathbf{M}_{mn}\}$.

- Show that U is a subspace of $\mathbb{R}^m$.
- Show that $U = \mathbb{R}^m$ if $\mathbf{x} \neq \mathbf{0}$.

Exercise 6.2.6 Write each of the following as a linear combination of $x+1$, x^2+x , and x^2+2 .

- $x^2 + 3x + 2$
- $2x^2 - 3x + 1$
- $x^2 + 1$
- x

Exercise 6.2.7 Determine whether $\mathbf{v}$ lies in $\text{span}\{\mathbf{u}, \mathbf{w}\}$ in each case.

- $\mathbf{v} = 3x^2 - 2x - 1$; $\mathbf{u} = x^2 + 1$, $\mathbf{w} = x + 2$
- $\mathbf{v} = x$; $\mathbf{u} = x^2 + 1$, $\mathbf{w} = x + 2$
- $\mathbf{v} = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}$; $\mathbf{u} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$
- $\mathbf{v} = \begin{bmatrix} 1 & -4 \\ 5 & 3 \end{bmatrix}$; $\mathbf{u} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$

Exercise 6.2.8 Which of the following functions lie in $\text{span}\{\cos^2 x, \sin^2 x\}$? (Work in $\mathbf{F}[0, \pi]$.)

- $\cos 2x$
- 1
- x^2
- $1 + x^2$

Exercise 6.2.9

- Show that $\mathbb{R}^3$ is spanned by $\{(1, 0, 1), (1, 1, 0), (0, 1, 1)\}$.
- Show that $\mathbf{P}_2$ is spanned by $\{1 + 2x^2, 3x, 1 + x\}$.
- Show that $\mathbf{M}_{22}$ is spanned by $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right\}$.

Exercise 6.2.10 If X and Y are two sets of vectors in a vector space V , and if $X \subseteq Y$, show that $\text{span } X \subseteq \text{span } Y$.

Exercise 6.2.11 Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ denote vectors in a vector space V . Show that:

- $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{span}\{\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{w}, \mathbf{v} + \mathbf{w}\}$
- $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{span}\{\mathbf{u} - \mathbf{v}, \mathbf{u} + \mathbf{w}, \mathbf{w}\}$

Exercise 6.2.12 Show that

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{0}\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

holds for any set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

Exercise 6.2.13 If X and Y are nonempty subsets of a vector space V such that $\text{span } X = \text{span } Y = V$, must there be a vector common to both X and Y ? Justify your answer.

Exercise 6.2.14 Is it possible that $\{(1, 2, 0), (1, 1, 1)\}$ can span the subspace $U = \{(a, b, 0) \mid a \text{ and } b \text{ in } \mathbb{R}\}$?

Exercise 6.2.15 Describe $\text{span}\{\mathbf{0}\}$.

Exercise 6.2.16 Let $\mathbf{v}$ denote any vector in a vector space V . Show that $\text{span}\{\mathbf{v}\} = \text{span}\{a\mathbf{v}\}$ for any $a \neq 0$.

Exercise 6.2.17 Determine all subspaces of $\mathbb{R}\mathbf{v}$ where $\mathbf{v} \neq \mathbf{0}$ in some vector space V .

Exercise 6.2.18 Suppose $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. If $\mathbf{u} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n$ where the a_i are in $\mathbb{R}$ and $a_1 \neq 0$, show that $V = \text{span}\{\mathbf{u}, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

Exercise 6.2.19 If $\mathbf{M}_{nn} = \text{span}\{A_1, A_2, \dots, A_k\}$, show that $\mathbf{M}_{nn} = \text{span}\{A_1^T, A_2^T, \dots, A_k^T\}$.

Exercise 6.2.20 If $P_n = \text{span}\{p_1(x), p_2(x), \dots, p_k(x)\}$ and a is in $\mathbb{R}$, show that $p_i(a) \neq 0$ for some i .

Exercise 6.2.21 Let U be a subspace of a vector space V .

- If au is in U where $a \neq 0$, show that u is in U .
- If u and $u + v$ are in U , show that v is in U .

Exercise 6.2.22 Let U be a nonempty subset of a vector space V . Show that U is a subspace of V if and only if $u_1 + au_2$ lies in U for all u_1 and u_2 in U and all a in $\mathbb{R}$.

Exercise 6.2.23 Let $U = \{p(x) \text{ in } P \mid p(3) = 0\}$ be the set in Example 6.2.4. Use the factor theorem (see Section 6.5) to show that U consists of multiples of $x - 3$; that is, show that $U = \{(x - 3)q(x) \mid q(x) \in P\}$. Use this to show that U is a subspace of P .

Exercise 6.2.24 Let $A_1, A_2, \dots, A_m$ denote $n \times n$ matrices. If $\mathbf{0} \neq \mathbf{y} \in \mathbb{R}^n$ and $A_1\mathbf{y} = A_2\mathbf{y} = \dots = A_m\mathbf{y} = \mathbf{0}$, show that $\{A_1, A_2, \dots, A_m\}$ cannot span M_{nn} .

Exercise 6.2.25 Let $\{v_1, v_2, \dots, v_n\}$ and $\{u_1, u_2, \dots, u_n\}$ be sets of vectors in a vector space, and let

$$X = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \quad Y = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

as in Exercise 6.1.18.

- Show that $\text{span}\{v_1, \dots, v_n\} \subseteq \text{span}\{u_1, \dots, u_n\}$ if and only if $AY = X$ for some $n \times n$ matrix A .
- If $X = AY$ where A is invertible, show that $\text{span}\{v_1, \dots, v_n\} = \text{span}\{u_1, \dots, u_n\}$.

Exercise 6.2.26 If U and W are subspaces of a vector space V , let $U \cup W = \{v \mid v \text{ is in } U \text{ or } v \text{ is in } W\}$. Show that $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

Exercise 6.2.27 Show that P cannot be spanned by a finite set of polynomials.

6.3 Linear Independence and Dimension

Definition 6.4 Linear Independence and Dependence

As in $\mathbb{R}^n$, a set of vectors $\{v_1, v_2, \dots, v_n\}$ in a vector space V is called **linearly independent** (or **simply independent**) if it satisfies the following condition:

$$\text{If } s_1 v_1 + s_2 v_2 + \dots + s_n v_n = \mathbf{0}, \quad \text{then } s_1 = s_2 = \dots = s_n = 0.$$

A set of vectors that is not linearly independent is said to be **linearly dependent** (or **simply dependent**).

The **trivial linear combination** of the vectors $v_1, v_2, \dots, v_n$ is the one with every coefficient zero:

$$0v_1 + 0v_2 + \dots + 0v_n$$

This is obviously one way of expressing $\mathbf{0}$ as a linear combination of the vectors $v_1, v_2, \dots, v_n$, and they are linearly independent when it is the *only* way.

V , this shows that V is a vector space (being a subspace of M_{22}). Now a matrix A is symmetric when entries directly across the main diagonal are equal, so each 2×2 symmetric matrix has the form

$$\begin{bmatrix} a & c \\ c & b \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + c \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Hence the set $B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ spans V , and the reader can verify that B is linearly independent. Thus B is a basis of V , so $\dim V = 3$.

It is frequently convenient to alter a basis by multiplying each basis vector by a nonzero scalar. The next example shows that this always produces another basis. The proof is left as Exercise 6.3.22.

Example 6.3.12

Let $B = \{v_1, v_2, \dots, v_n\}$ be nonzero vectors in a vector space V . Given nonzero scalars $a_1, a_2, \dots, a_n$, write $D = \{a_1 v_1, a_2 v_2, \dots, a_n v_n\}$. If B is independent or spans V , the same is true of D . In particular, if B is a basis of V , so also is D .

Exercises for 6.3

Exercise 6.3.1 Show that each of the following sets of vectors is independent.

a. $\{1+x, 1-x, x+x^2\}$ in P_2

b. $\{x^2, x+1, 1-x-x^2\}$ in P_2

c. $\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right\}$
in M_{22}

d. $\left\{ \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right\}$
in M_{22}

Exercise 6.3.2 Which of the following subsets of V are independent?

a. $V = P_2; \{x^2+1, x+1, x\}$

b. $V = P_2; \{x^2-x+3, 2x^2+x+5, x^2+5x+1\}$

c. $V = M_{22}; \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

d. $V = M_{22}; \left\{ \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \right\}$

e. $V = F[1, 2]; \left\{ \frac{1}{x}, \frac{1}{x^2}, \frac{1}{x^3} \right\}$

f. $V = F[0, 1]; \left\{ \frac{1}{x^2+x-6}, \frac{1}{x^2-5x+6}, \frac{1}{x^2-9} \right\}$

Exercise 6.3.3 Which of the following are independent in $F[0, 2\pi]$?

a. $\{\sin^2 x, \cos^2 x\}$

b. $\{1, \sin^2 x, \cos^2 x\}$

c. $\{x, \sin^2 x, \cos^2 x\}$

Exercise 6.3.4 Find all values of a such that the following are independent in $\mathbb{R}^3$.

a. $\{(1, -1, 0), (a, 1, 0), (0, 2, 3)\}$

- b. $\{(2, a, 1), (1, 0, 1), (0, 1, 3)\}$

Exercise 6.3.5 Show that the following are bases of the space V indicated.

- a. $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}; V = \mathbb{R}^3$
 b. $\{(-1, 1, 1), (1, -1, 1), (1, 1, -1)\}; V = \mathbb{R}^3$
 c. $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}; V = \mathbf{M}_{22}$
 d. $\{1+x, x+x^2, x^2+x^3, x^3\}; V = \mathbf{P}_3$

Exercise 6.3.6 Exhibit a basis and calculate the dimension of each of the following subspaces of $\mathbf{P}_2$.

- a. $\{a(1+x) + b(x+x^2) \mid a \text{ and } b \text{ in } \mathbb{R}\}$
 b. $\{a + b(x+x^2) \mid a \text{ and } b \text{ in } \mathbb{R}\}$
 c. $\{p(x) \mid p(1) = 0\}$
 d. $\{p(x) \mid p(x) = p(-x)\}$

Exercise 6.3.7 Exhibit a basis and calculate the dimension of each of the following subspaces of $\mathbf{M}_{22}$.

- a. $\{A \mid A^T = -A\}$
 b. $\left\{ A \mid A \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} A \right\}$
 c. $\left\{ A \mid A \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\}$
 d. $\left\{ A \mid A \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} A \right\}$

Exercise 6.3.8 Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and define $U = \{X \mid X \in \mathbf{M}_{22} \text{ and } AX = X\}$.

- a. Find a basis of U containing A .
 b. Find a basis of U not containing A .

Exercise 6.3.9 Show that the set $\mathbb{C}$ of all complex numbers is a vector space with the usual operations, and find its dimension.

Exercise 6.3.10

- a. Let V denote the set of all 2×2 matrices with equal column sums. Show that V is a subspace of $\mathbf{M}_{22}$, and compute $\dim V$.
 b. Repeat part (a) for 3×3 matrices.
 c. Repeat part (a) for $n \times n$ matrices.

Exercise 6.3.11

- a. Let $V = \{(x^2 + x + 1)p(x) \mid p(x) \text{ in } \mathbf{P}_2\}$. Show that V is a subspace of $\mathbf{P}_4$ and find $\dim V$. [Hint: If $f(x)g(x) = 0$ in $\mathbf{P}$, then $f(x) = 0$ or $g(x) = 0$.]
 b. Repeat with $V = \{(x^2 - x)p(x) \mid p(x) \text{ in } \mathbf{P}_3\}$, a subset of $\mathbf{P}_5$.
 c. Generalize.

Exercise 6.3.12 In each case, either prove the assertion or give an example showing that it is false.

- a. Every set of four nonzero polynomials in $\mathbf{P}_3$ is a basis.
 b. $\mathbf{P}_2$ has a basis of polynomials $f(x)$ such that $f(0) = 0$.
 c. $\mathbf{P}_2$ has a basis of polynomials $f(x)$ such that $f(0) = 1$.
 d. Every basis of $\mathbf{M}_{22}$ contains a noninvertible matrix.
 e. No independent subset of $\mathbf{M}_{22}$ contains a matrix A with $A^2 = 0$.
 f. If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent then, $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \mathbf{0}$ for some a, b, c .
 g. $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent if $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \mathbf{0}$ for some a, b, c .
 h. If $\{\mathbf{u}, \mathbf{v}\}$ is independent, so is $\{\mathbf{u}, \mathbf{u} + \mathbf{v}\}$.
 i. If $\{\mathbf{u}, \mathbf{v}\}$ is independent, so is $\{\mathbf{u}, \mathbf{v}, \mathbf{u} + \mathbf{v}\}$.
 j. If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent, so is $\{\mathbf{u}, \mathbf{v}\}$.
 k. If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent, so is $\{\mathbf{u} + \mathbf{w}, \mathbf{v} + \mathbf{w}\}$.
 l. If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent, so is $\{\mathbf{u} + \mathbf{v} + \mathbf{w}\}$.

- m. If $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$ then $\{\mathbf{u}, \mathbf{v}\}$ is dependent if and only if one is a scalar multiple of the other.
- n. If $\dim V = n$, then no set of more than n vectors can be independent.
- o. If $\dim V = n$, then no set of fewer than n vectors can span V .

Exercise 6.3.13 Let $A \neq 0$ and $B \neq 0$ be $n \times n$ matrices, and assume that A is symmetric and B is skew-symmetric (that is, $B^T = -B$). Show that $\{A, B\}$ is independent.

Exercise 6.3.14 Show that every set of vectors containing a dependent set is again dependent.

Exercise 6.3.15 Show that every nonempty subset of an independent set of vectors is again independent.

Exercise 6.3.16 Let f and g be functions on $[a, b]$, and assume that $f(a) = 1 = g(b)$ and $f(b) = 0 = g(a)$. Show that $\{f, g\}$ is independent in $\mathbf{F}[a, b]$.

Exercise 6.3.17 Let $\{A_1, A_2, \dots, A_k\}$ be independent in $\mathbf{M}_{mn}$, and suppose that U and V are invertible matrices of size $m \times m$ and $n \times n$, respectively. Show that $\{UA_1V, UA_2V, \dots, UA_kV\}$ is independent.

Exercise 6.3.18 Show that $\{\mathbf{v}, \mathbf{w}\}$ is independent if and only if neither $\mathbf{v}$ nor $\mathbf{w}$ is a scalar multiple of the other.

Exercise 6.3.19 Assume that $\{\mathbf{u}, \mathbf{v}\}$ is independent in a vector space V . Write $\mathbf{u}' = a\mathbf{u} + b\mathbf{v}$ and $\mathbf{v}' = c\mathbf{u} + d\mathbf{v}$, where a, b, c , and d are numbers. Show that $\{\mathbf{u}', \mathbf{v}'\}$ is independent if and only if the matrix $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$ is invertible. [Hint: Theorem 2.4.5.]

Exercise 6.3.20 If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent and $\mathbf{w}$ is not in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, show that:

- a. $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent.
- b. $\{\mathbf{v}_1 + \mathbf{w}, \mathbf{v}_2 + \mathbf{w}, \dots, \mathbf{v}_k + \mathbf{w}\}$ is independent.

Exercise 6.3.21 If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent, show that $\{\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \dots, \mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_k\}$ is also independent.

Exercise 6.3.22 Prove Example 6.3.12.

Exercise 6.3.23 Let $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{z}\}$ be independent. Which of the following are dependent?

- a. $\{\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w}, \mathbf{w} - \mathbf{u}\}$

- b. $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u}\}$
- c. $\{\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w}, \mathbf{w} - \mathbf{z}, \mathbf{z} - \mathbf{u}\}$
- d. $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{z}, \mathbf{z} + \mathbf{u}\}$

Exercise 6.3.24 Let U and W be subspaces of V with bases $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $\{\mathbf{w}_1, \mathbf{w}_2\}$ respectively. If U and W have only the zero vector in common, show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1, \mathbf{w}_2\}$ is independent.

Exercise 6.3.25 Let $\{p, q\}$ be independent polynomials. Show that $\{p, q, pq\}$ is independent if and only if $\deg p \geq 1$ and $\deg q \geq 1$.

Exercise 6.3.26 If z is a complex number, show that $\{z, z^2\}$ is independent if and only if z is not real.

Exercise 6.3.27 Let $B = \{A_1, A_2, \dots, A_n\} \subseteq \mathbf{M}_{mn}$, and write $B' = \{A_1^T, A_2^T, \dots, A_n^T\} \subseteq \mathbf{M}_{nm}$. Show that:

- a. B is independent if and only if B' is independent.
- b. B spans $\mathbf{M}_{mn}$ if and only if B' spans $\mathbf{M}_{nm}$.

Exercise 6.3.28 If $V = \mathbf{F}[a, b]$ as in Example 6.1.7, show that the set of constant functions is a subspace of dimension 1 (f is constant if there is a number c such that $f(x) = c$ for all x).

Exercise 6.3.29

- a. If U is an invertible $n \times n$ matrix and $\{A_1, A_2, \dots, A_{mn}\}$ is a basis of $\mathbf{M}_{mn}$, show that $\{A_1U, A_2U, \dots, A_{mn}U\}$ is also a basis.
- b. Show that part (a) fails if U is not invertible. [Hint: Theorem 2.4.5.]

Exercise 6.3.30 Show that $\{(a, b), (a_1, b_1)\}$ is a basis of $\mathbb{R}^2$ if and only if $\{a + bx, a_1 + b_1x\}$ is a basis of $\mathbf{P}_1$.

Exercise 6.3.31 Find the dimension of the subspace $\text{span}\{1, \sin^2 \theta, \cos 2\theta\}$ of $\mathbf{F}[0, 2\pi]$.

Exercise 6.3.32 Show that $\mathbf{F}[0, 1]$ is not finite dimensional.

Exercise 6.3.33 If U and W are subspaces of V , define their intersection $U \cap W$ as follows:

$$U \cap W = \{\mathbf{v} \mid \mathbf{v} \text{ is in both } U \text{ and } W\}$$

- a. Show that $U \cap W$ is a subspace contained in U and W .

- b. Show that $U \cap W = \{0\}$ if and only if $\{u, w\}$ is independent for any nonzero vectors u in U and w in W .
- c. If B and D are bases of U and W , and if $U \cap W = \{0\}$, show that $B \cup D = \{v \mid v \text{ is in } B \text{ or } D\}$ is independent.

Exercise 6.3.34 If U and W are vector spaces, let $V = \{(u, w) \mid u \text{ in } U \text{ and } w \text{ in } W\}$.

- a. Show that V is a vector space if $(u, w) + (u_1, w_1) = (u + u_1, w + w_1)$ and $a(u, w) = (au, aw)$.
- b. If $\dim U = m$ and $\dim W = n$, show that $\dim V = m + n$.
- c. If $V_1, \dots, V_m$ are vector spaces, let

$$\begin{aligned} V &= V_1 \times \cdots \times V_m \\ &= \{(v_1, \dots, v_m) \mid v_i \in V_i \text{ for each } i\} \end{aligned}$$

denote the space of n -tuples from the V_i with componentwise operations (see Exercise 6.1.17). If $\dim V_i = n_i$ for each i , show that $\dim V = n_1 + \cdots + n_m$.

Exercise 6.3.35 Let D_n denote the set of all functions f from the set $\{1, 2, \dots, n\}$ to $\mathbb{R}$.

- a. Show that D_n is a vector space with pointwise addition and scalar multiplication.
- b. Show that $\{S_1, S_2, \dots, S_n\}$ is a basis of D_n where, for each $k = 1, 2, \dots, n$, the function S_k is defined by $S_k(k) = 1$, whereas $S_k(j) = 0$ if $j \neq k$.

Exercise 6.3.36 A polynomial $p(x)$ is called **even** if $p(-x) = p(x)$ and **odd** if $p(-x) = -p(x)$. Let E_n and O_n denote the sets of even and odd polynomials in P_n .

- a. Show that E_n is a subspace of P_n and find $\dim E_n$.
- b. Show that O_n is a subspace of P_n and find $\dim O_n$.

Exercise 6.3.37 Let $\{v_1, \dots, v_n\}$ be independent in a vector space V , and let A be an $n \times n$ matrix. Define $u_1, \dots, u_n$ by

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = A \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

(See Exercise 6.1.18.) Show that $\{u_1, \dots, u_n\}$ is independent if and only if A is invertible.

6.4 Finite Dimensional Spaces

Up to this point, we have had no guarantee that an arbitrary vector space *has* a basis—and hence no guarantee that one can speak *at all* of the dimension of V . However, Theorem 6.4.1 will show that any space that is spanned by a finite set of vectors has a (finite) basis: The proof requires the following basic lemma, of interest in itself, that gives a way to enlarge a given independent set of vectors.

Lemma 6.4.1: Independent Lemma

Let $\{v_1, v_2, \dots, v_k\}$ be an independent set of vectors in a vector space V . If $u \in V$ but⁵ $u \notin \text{span}\{v_1, v_2, \dots, v_k\}$, then $\{u, v_1, v_2, \dots, v_k\}$ is also independent.

Proof. Let $tu + t_1v_1 + t_2v_2 + \cdots + t_kv_k = 0$; we must show that all the coefficients are zero. First, $t = 0$ because, otherwise, $u = -\frac{t_1}{t}v_1 - \frac{t_2}{t}v_2 - \cdots - \frac{t_k}{t}v_k$ is in $\text{span}\{v_1, v_2, \dots, v_k\}$, contrary to our assumption.

⁵If X is a set, we write $a \in X$ to indicate that a is an element of the set X . If a is not an element of X , we write $a \notin X$.